

Experiment No. 6

Name	Roll No
Palak Chanchlani	05
Sonam Chhabaidiya	09
Mahi Jodhani	21

AIM: Interfacing and simulation with Arduino a smart RFID-based door lock system

Introduction: An RFID-based door lock system is a secure and contactless authentication mechanism that uses RFID tags or cards to grant or deny access. This project focuses on building such a system using Arduino Mega, RFID RC522 module, Servo Motor, and I2C LCD. Unlike traditional locking systems that require keys or keypads, this system provides access based on the unique ID of the RFID card presented.

Component Required:

- Arduino Uno/Mega (Controller): Acts as the main brain of the project, controlling all components and processing RFID authentication.
- RFID Module RC522: Used to read the UID from RFID cards or tags when they are brought near the scanner.
- RFID Tags/Cards: Each tag contains a unique identifier (UID) used to verify and grant or deny access.
- Servo Motor (Locking Mechanism): Physically rotates to unlock or lock the door when access is granted.
- I2C LCD Display (16x2): Provides real-time visual feedback to the user like "Access Granted" or "Access Denied."
- Buzzer (Optional): Produces sound feedback indicating success or failure of access attempt.

- g. Breadboard and Jumper Wires: Used to build the circuit and make connections between all components without soldering.
- h. Power Source (USB/5V/9V Battery): Provides necessary power to run the Arduino and other connected components.

Connections:

➤ RFID RC522 Module:

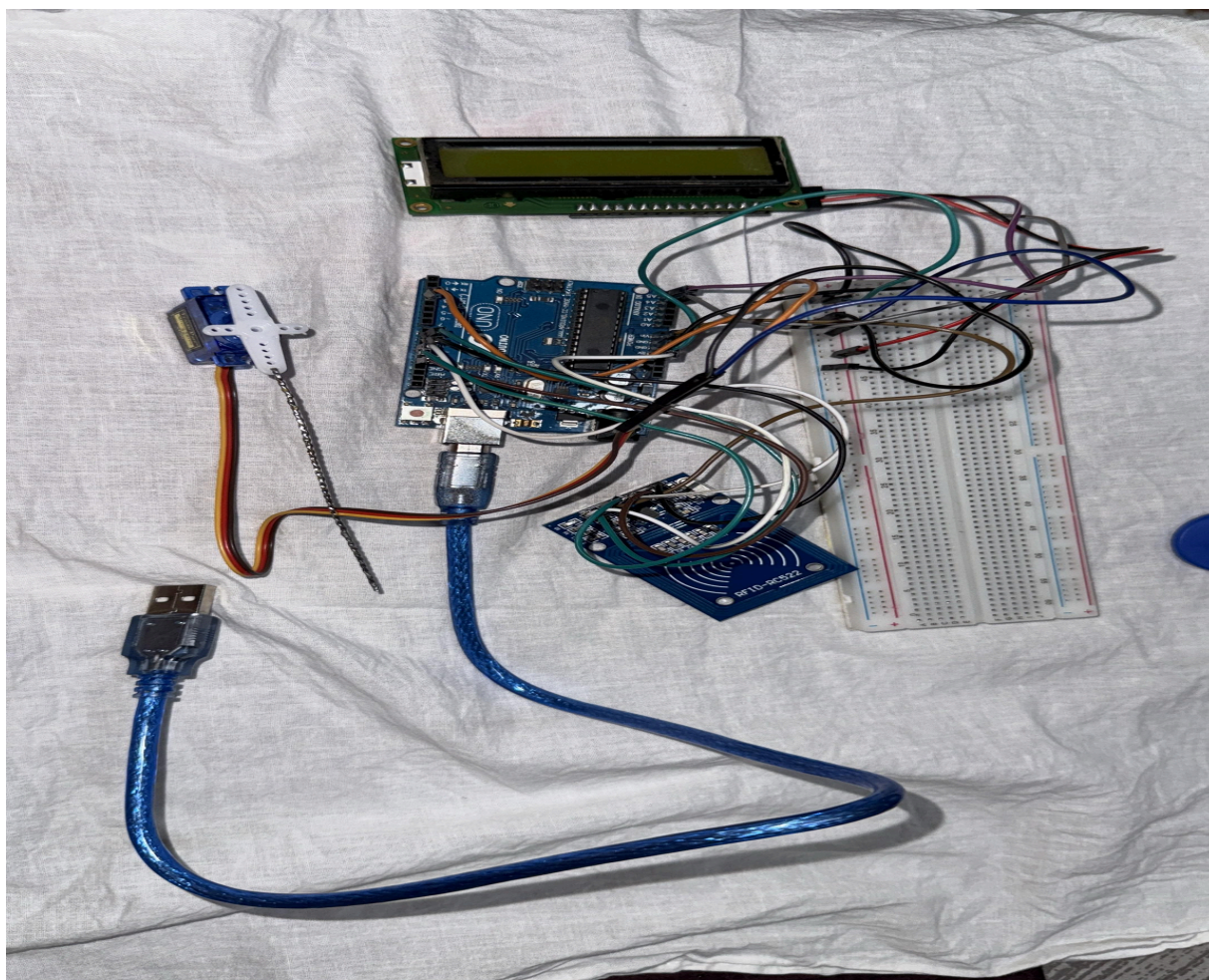
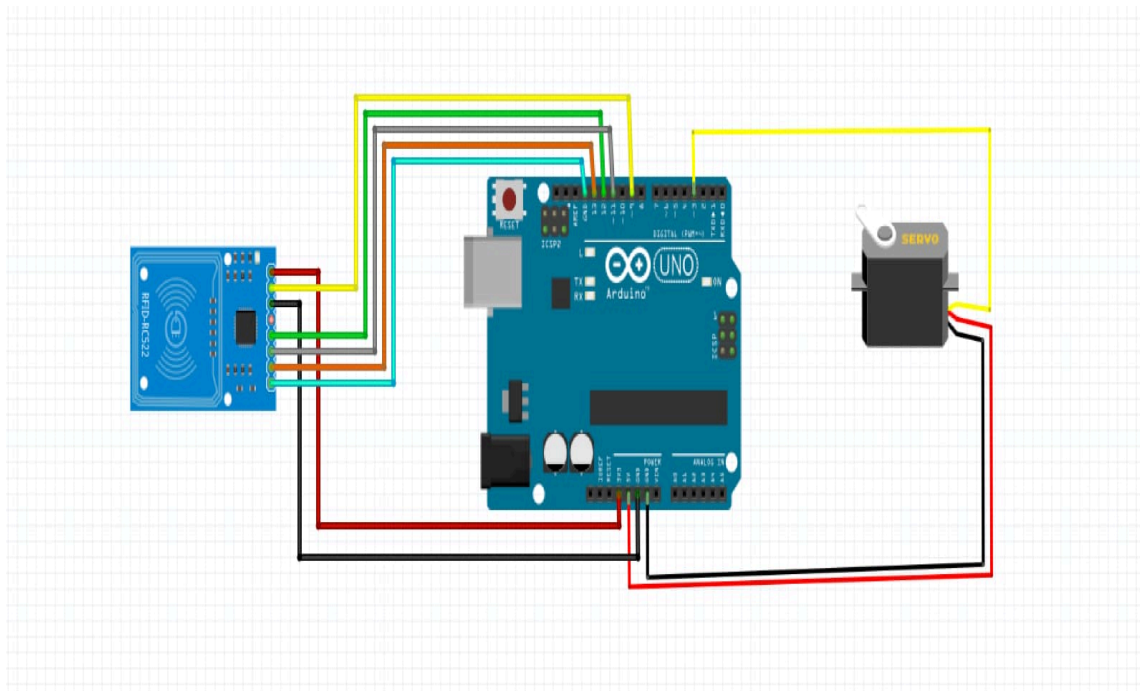
- SDA → Arduino Mega Pin 10 (SS)
- SCK → Arduino Mega Pin 52
- MOSI → Arduino Mega Pin 51
- MISO → Arduino Mega Pin 50
- RST → Arduino Mega Pin 9
- GND → GND
- VCC → 3.3V

➤ Servo Motor:

- Signal → Arduino Mega Pin 3
- VCC → 5V
- GND → GND

➤ I2C LCD Display (16x2):

- SDA → Arduino Mega Pin 20
- SCL → Arduino Mega Pin 21
- VCC → 5V



Working Principle:

The RFID-based door lock system operates on a simple authentication mechanism using RFID cards and an Arduino microcontroller. When the system is powered on, the LCD immediately displays the message “**Scan Your Card**”, indicating that it is ready to accept an RFID input. The RC522 RFID reader continuously monitors for the presence of a card. Once a card is brought near the reader, it reads the unique identification number (UID) embedded in the card.

This UID is then compared with a **predefined authorized UID** stored in the Arduino’s code. If a match is found, the system recognizes the card as valid and proceeds to unlock the door by rotating the **servo motor to 90°**. Simultaneously, the LCD displays “**Access Granted**” as a confirmation to the user. After a delay of approximately 3 seconds, the servo motor returns to its original position (0°), locking the door again, and the LCD resets to “**Scan Your Card**”. If an unauthorized UID is scanned, the LCD shows “**Access Denied**”, and no change is made to the servo motor position. This simple, user-friendly flow ensures a secure and contactless entry mechanism.

Software Implementation:

The software for this project was written using the Arduino IDE and makes use of several standard libraries such as MFRC522 for reading RFID cards, Servo for controlling the motor, and LiquidCrystal_I2C for displaying messages on the LCD.

When the system starts, it initializes the RFID module, LCD, and servo motor. In the main loop, the RFID reader continuously waits for a card to be scanned. Once a card is detected, the system reads its **UID** and compares it with a **predefined authorized UID**. If the UID matches, the servo rotates to unlock the door and the LCD displays “Access Granted.” Otherwise, it shows “Access Denied.” After a delay, the system resets to wait for the next scan.

This simple loop-based logic provides real-time feedback and control over the lock mechanism

Arduino Code:

```
#include <SPI.h>

#include <MFRC522.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include <Servo.h>

#define SS_PIN 10

#define RST_PIN 9
```

```

MFRC522 rfid(SS_PIN, RST_PIN);

LiquidCrystal_I2C lcd(0x27, 16, 2);

Servo lockServo;

byte authorizedUID[] = {0xA1, 0xB2, 0xC3, 0xD4};

void setup() {

    Serial.begin(9600);

    SPI.begin();

    rfid.PCD_Init();

    lcd.begin();

    lcd.backlight();

    lcd.setCursor(0, 0);

    lcd.print("Scan Your Card");

    lockServo.attach(3);

    lockServo.write(0); // Locked

}

void loop() {

    if (!rfid.PICC_IsNewCardPresent() || !rfid.PICC_ReadCardSerial()) return;

    byte* id = rfid.uid.uidByte;

    bool authorized = true;

    for (byte i = 0; i < rfid.uid.size; i++) {

        if (id[i] != authorizedUID[i]) {

            authorized = false;

            Break; } }

    lcd.clear();

```

```

if (authorized) {

  lcd.setCursor(0, 0);

  lcd.print("Access Granted");

  lockServo.write(90);

  delay(3000);

  lockServo.write(0);

} else {

  lcd.setCursor(0, 0);

  lcd.print("Access Denied");

  delay(2000);

}

lcd.clear();

lcd.setCursor(0, 0);

lcd.print("Scan Your Card");

rfid.PICC_HaltA();

rfid.PCD_StopCrypto1();

}

```

Communication:

The project uses multiple communication protocols to establish proper interaction between hardware components and the Arduino Mega. Each module is connected using specific protocols tailored to its function and hardware compatibility.

1. SPI (Serial Peripheral Interface) — RFID Reader: The RC522 RFID module uses the SPI protocol to communicate with the Arduino. SPI is a high-speed synchronous serial communication method that connects the reader via the MOSI, MISO, SCK, SDA, and RST pins. It allows the Arduino to read the UID from the RFID card with minimal delay, making the system responsive.
2. PWM (Pulse Width Modulation) — Servo Motor: The servo motor is controlled using PWM signals sent from a digital pin (Pin 3). PWM allows the Arduino to control the angle

of the motor by varying the duty cycle of the signal. When access is granted, the servo rotates to 90°, and then after a short delay, it returns to 0° (locked position).

3. I2C (Inter-Integrated Circuit) — LCD Display: The I2C LCD (16x2) uses two wires — SDA (data line) and SCL (clock line) — to communicate with the Arduino. This communication is handled via the `Wire` library. I2C reduces the number of required pins and enables real-time feedback display, such as “Scan Your Card,” “Access Granted,” or “Access Denied.”
4. Digital Signal — Buzzer (Optional): The buzzer is controlled by a digital pin (e.g., Pin 4). It operates by receiving HIGH/LOW signals to produce sound. When access is granted or denied, short tones provide auditory feedback to the user.

Conclusion:

The RFID-based door lock system developed in this project successfully fulfills the requirement for a secure, contactless, and low-cost access control mechanism. By using RFID technology in place of traditional keys or keypad systems, the project offers a modern and efficient solution that is particularly suited for homes, labs, offices, and academic environments.

The system is designed with simplicity, scalability, and reliability in mind. It uses readily available components and can be enhanced further by adding EEPROM for saving multiple UIDs, integrating with GSM or IoT modules for remote monitoring, or adding logs for access events. The project not only demonstrates a clear understanding of embedded systems and electronics but also provides a strong foundation for developing more advanced security systems in the future.